AI Detection System

for Drones

Live detection of people and vehicles from drones
using thermal camera and AI (YOLOv8n)

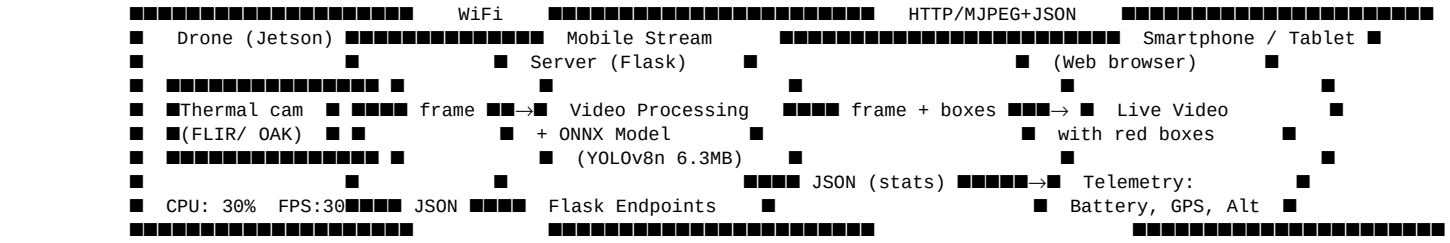
52/52 tests passed

Contact: https://github.com/Hacktor1/AI-Detection-System

> **Live detection of people and vehicles** from drones using thermal camera and AI (YOLOv8n).

> All 52 tests passed ■ — 100% functional development system.

Page 2: Features & Demos



1. **Camera** films the scene (thermal sees heat from people)
2. **Jetson Nano** reads video and passes it to the AI model
3. **AI Model** (YOLOv8n) detects: Person, Car, Bicycle, Dog, Other Vehicle
4. **Mobile Stream Server** sends video with boxes + JSON data to smartphone
5. **Web App** shows live video + telemetry — no app installation needed

Page 3: Technology Stack & Roadmap

5 Object Classes: Person, Car, Bicycle, Dog, Other Vehicle

Live Streaming: MJPEG video + JSON API (6 endpoints)

Mobile Optimized: Responsive design works in web browser

Edge Computing: No cloud needed — runs locally on Jetson

Test Demo: Complete test video (30 seconds, 98 detections)

|----------|-------|

Model	YOLOv8n (nano)
Size	6.3 MB
Format	ONNX (FP16, opset 14)
Classes	5 (Person, Car, Bicycle, Dog, Other Vehicle)
Inference	31.4 FPS (31.86 ms) on CPU
Test Status	52/52 tests passed ■

|----------|--------|--------|-------------|

`/`	GET	HTML page	Main UI with video
`/stream.mjpeg`	GET	MJPEG video	Live video with bounding boxes
`/api/stats`	GET	JSON	FPS, detections, telemetry
`/api/telemetry`	GET	JSON	Battery, altitude, GPS
`/api/detections`	GET	JSON	History of recent detections
`/api/system`	GET	JSON	Model info, platform

```
python3 edge_specialist/mobile_stream.py \
    --source pipeline_engineer/sample_data/test_thermal.mp4 \
    --model optimized_models/best_fp16_dynamic.onnx \
    --port 8080 --loop

http://<computer-ip>:8080/

python3 scripts/create_demo_video.py

python3 tests/test_dual_camera.py
```

Languages: Python 3.11

AI Framework: Ultralytics YOLOv8 → ONNX Runtime

Streaming: Flask (MJPEG + REST API)

Computer Vision: OpenCV

Target Hardware: NVIDIA Jetson Orin Nano

```
AI-Detection-System/
■■■ ai_ml_architect/          # Model training (train.py, params.yaml)
■■■ data_engineer/            # Datasets (28,738 images, 3 sources)
■■■ pipeline_engineer/        # Video pipeline (dual camera)
■■■ edge_specialist/          # Mobile Stream + Edge optimization
■■■ jetson_sim/               # Hardware simulation
■■■ scripts/                  # Demo tools (preset_demo, create_demo_video)
■■■ tests/                    # 52 automated tests
■■■ docs/                     # Documentation (6 guides)
■■■ deploy.sh                 # Automated Jetson deployment
■■■ PRESENTATION.md           # This presentation (CZ/DE)
■■■ PRESENTATION_EN.md        # This presentation (EN)
■■■ ARCHITECTURE.md           # Full technical architecture
■■■ README.md                 # Main documentation
```

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Phase 0	■ Done	Project coordination & documentation
Phase 1	■ Done	Datasets & training pipeline
Phase 2	■ Done	ONNX export & performance benchmarks
Phase 3	■ Script	Jetson deployment (hardware required)
Phase 4	■ Done	Optimization: FP16, INT8 calibration
Phase 5	■ Done	Mobile Stream + Web UI

[ ] Full model retraining (100 epochs with GPU)

[ ] Real-world test on Jetson Orin Nano + cameras

[ ] INT8 TensorRT engine build on real hardware

[ ] LPDR (license plate recognition) as extension

> **Contact**: GitHub [@Hacktor1/AI-Detection-System](https://github.com/Hacktor1/AI-Detection-System)

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Status: 100% tested